

Predictive Modelling of COVID-19 In-Hospital Mortality Using Admission Variables: A Responsible-ML Pipeline Toward Long COVID Risk Assessment

Durga Prasad Narsing

School of Computing, Dublin City University
Glasnevin, Dublin 9, Ireland

durgaprasad.narsing2@mail.dcu.ie

Abstract—Severe COVID-19 outcomes, including in-hospital mortality, place a major burden on health systems, and identifying high-risk patients at the time of hospital admission supports earlier intervention. Reviewing existing predictive models shows that calibration is seldom reported, the fairness of the models over different demographic subgroups is not always explored, post-admission variables such as intubation status are too commonly used as predictors (target leakage), individual-level explanations are limited or entirely missing, and the resulting models are seldom delivered as a working prototype. We cover all five gaps through a responsible-ML pipeline with 13 phases over 220,218 patients with confirmed COVID-19 cases from the Mexican Government surveillance dataset, all without a data leak. The prediction target is in-hospital mortality, a directly labelled severe outcome, consistent with previous work [1][2]. Five model families were compared using nine admission-time features. The Logistic Regression yielded an AUC of 0.888 (95% CI 0.884-0.892), and the DeLong test showed no statistically significant difference in AUC between the logistic model and the complex ensembles ($p > 0.05$). The Brier score reduced from 0.135 to 0.078 with isotonic calibration, and count-weighted ECE from 0.195 to 0.002. Across the eight subgroups the model is calibrated equitably ($ECE < 0.015$), and TPR disparity is further reduced by group-specific thresholds (TPR disparity < 0.045). SHAP analysis of the deployed model identified pneumonia (39.9%) and age (33.1%) as dominant predictors. A Streamlit research prototype delivers calibrated risk scores with model-consistent explanations. Extending this pipeline to PASC-labelled cohorts for direct Long COVID prediction is identified as the key next step.

Index Terms—Long COVID, PASC, COVID-19 mortality, machine learning, calibration, DeLong test, SHAP, fairness audit, responsible AI, clinical decision support, Streamlit.

I. INTRODUCTION

One of the most striking public-health issues to emerge from the pandemic is Post-COVID-19 Condition (PASC), or Long COVID. As reported in a recent systematic review and meta-analysis [3], a portion of individuals infected with SARS-CoV-2 develop symptoms following the acute illness (fatigue, dyspnoea, cognitive dysfunction and cardiovascular sequelae) that persist months or years after the initial infection, termed PASC [4]. According to national surveillance data, in the U.S. 6.9% of adults have had at least one symptom of Long COVID at some point, and 3.4% still have current symptoms [5]. Older people, females and those

with chronic diseases seem especially affected. The widespread and highly varied nature of Long COVID is obvious, and there is a clear need for clinically accepted tools that can identify high-risk populations at the time of admission, before the window for early intervention closes.

While there has been significant growth in the use of predictive modelling tools, these tools still tend to share the same methodological drawbacks. Five common gaps emerge in the literature from the systematic reviews [6][7]: calibration is rarely reported, subgroup fairness is seldom examined, post-admission variables (intubation, ICU status) leak target information, per-patient explanations are missing, and models are rarely delivered as a working prototype. There is also an open question to test formally, within the class of high-risk problems with predominantly binary features: does a complex ensemble outperform logistic regression by a significant margin on clinical datasets? [8]

This study addresses all five gaps with an integrated 13-phase end-to-end responsible-ML pipeline. We use confirmed COVID-19 cases from the Mexican Government COVID-19 surveillance dataset ($n=220,218$), restricted to admission-day features; in-hospital mortality is the prediction target, a distinct and directly labelled severe outcome. Long COVID is a separate post-acute condition; predicting it directly would require PASC follow-up labels, and doing so is the intended extension of this pipeline (Section VI). This target choice is consistent with previous work [1, 2]. RQ1 - do the 9 admission features give clinically useful discrimination and calibration? RQ2 - by the DeLong test, do complex ensembles perform significantly better than Logistic Regression? RQ3 - are predictions equitable and calibrated across demographic subgroups? RQ4 - can predictions be delivered in an interpretable format via a functional prototype?

A. Contribution

Three contributions come from this work. The first is empirical: using DeLong's test on 220,218 patients, we show Logistic Regression's AUC is not significantly different from Random Forest, Gradient Boosting, XGBoost and a Stacking ensemble (all $p > 0.05$) - interpretability costs nothing in discrimination for low-dimensional binary features. The second is methodological: four of the five gaps - leakage-free preprocessing, probability calibration, subgroup fairness auditing and multi-layer explainability - are achieved within a single reproducible workflow. The third concerns responsible AI: we create a Model Card and Datasheet artefact for the model, implement group-specific threshold

mitigation, and embed a governance disclaimer directly in the Streamlit prototype, closing the fifth gap by delivering a functional research prototype.

B. Paper Structure

Section II reviews the literature and the five gaps. Section III describes the end-to-end pipeline. Section IV reports results across six evaluation dimensions. Section V discusses results and limitations. Section VI concludes.

II. RELATED WORK

The literature below spans five themes: risk-factor studies, ML prediction models, calibration, fairness, and explainability for Long COVID and COVID-19 mortality. We close by identifying the gap that motivated this study.

A. Risk-Factor Studies

Multiple studies have identified clinical factors associated with poor COVID-19 and Long COVID outcomes, including pneumonia, age, male sex, obesity and acute severity [1][9]. What is generally missing is the step from analysis to a validated, deployable tool. Menezes et al. [1] found strong associations for age ≥ 45 (OR 5.55) and fifteen or more acute symptoms (OR 6.02), but from only 108 patients at a single centre, limiting generalisability. Guzman-Esquivel et al. [9] reported hospitalisation (RR 2.1) and tachycardia (RR 10.4) as independent predictors, but without a calibrated model.

B. ML Prediction Models and the ML-vs-Regression Question

Turning to deployable models, the picture is thin. Jeffrey et al. [7] fitted a logistic regression on a Scottish cohort with five predictors without reporting calibration or fairness. Antony et al. [6] studied two million EHR patients with AUROC 0.75-0.80 but no fairness or calibration analysis. Interpretability was notably limited, while discrimination was good, in an ensemble of Super Learners applied to 57,563 patients by Dozier et al. [2]. Zang et al. [10] used regularised Cox models (C-index > 0.80) on the RECOVER EHR cohort without calibration or fairness auditing. Shakhovska et al. [11] reported AUC = 0.978 on only 122 patients (serious overfitting concerns). More generally, Bozkurt and Keskin [12] reached AUC 0.91 with gradient boosting; Cordelli et al. [13] predicted pulmonary Long COVID sequelae from clinical data; Kourmpanis et al. [14] achieved AUC 0.94; and Sun et al. [15] and Tang et al. [16] validated further mortality models. Most directly, Bjerre et al. [8] documented that complex ML models provide only minor benefits over regression, a conclusion we test formally here using DeLong statistics (RQ2).

C. Calibration and Clinical Utility

Strong discrimination on held-out data is not necessarily useful in the clinic; as Teeuw et al. [17] note, a crucial factor is how well predicted probabilities are calibrated. A clinician must be confident that a 70% risk score corresponds to a roughly 70% event rate. For this reason, calibration and Decision Curve Analysis are treated here as primary

deployment criteria rather than secondary checks, in contrast to most previous studies.

D. Fairness in Clinical ML

Although COVID-19 outcomes differ across age, sex and comorbidity groups, few Long COVID prediction papers consider subgroup fairness [2][7][10][18]. Two concepts are often conflated: discrimination (AUC) and calibration equity. We treat these as separate lenses and audit both, applying group-specific threshold mitigation where differences appear.

E. Explainability

Existing Long COVID models lack individual-level explanations [6][7][8]. Global feature-importance rankings are sometimes provided, but rarely with per-patient attributions or a practical bedside instrument. We combine three explainability layers - SHAP values, odds ratios and a clinical nomogram - into one framework and present them in a functional prototype.

F. Research Gap

No single study brings together leakage-free design, probability calibration, subgroup fairness auditing, and multi-layer explainability into one reproducible, deployable workflow. Table I maps the five recurring gaps to how this study addresses each.

TABLE I
Methodological Gaps in Prior Work vs This Study

Gap	Prior Studies	This Work
Calibration not reported	[2][7][10][11]	Brier + ECE; isotonic correction
No subgroup fairness audit	[2][7][10][18]	8 subgroups; threshold mitigation
Target leakage (intubation/ICU)	[11][18]	9 admission-time features only
No per-patient explainability	[6][7][8]	SHAP + OR + nomogram + web app
No deployable prototype	[1][7][9]	Streamlit app with live pipeline

III. METHODOLOGY

Our pipeline follows three core principles - no leakage, formal model selection, and responsible deployment. Five model families are benchmarked, the best is selected by a statistical test (not by visual ROC comparison), its probabilities are calibrated, and the final system exposes calibrated risk with multi-layer explanations through a prototype. The eight-stage architecture (Fig. 1) - patient inputs, leakage-free preprocessing, cost-sensitive benchmarking, DeLong-based model selection, isotonic calibration, the 13 KB deployed model, multi-layer outputs, and the Streamlit research prototype - is situated within the 13-phase ML lifecycle in Fig. 2. Post-admission variables (intubation, ICU admission) are excluded from the outset as target leakage, and the deployed model uses the identical impute $\rightarrow$ scale $\rightarrow$ predict chain at training and inference so no development/deployment shift can occur. Each component is detailed in the subsections below.

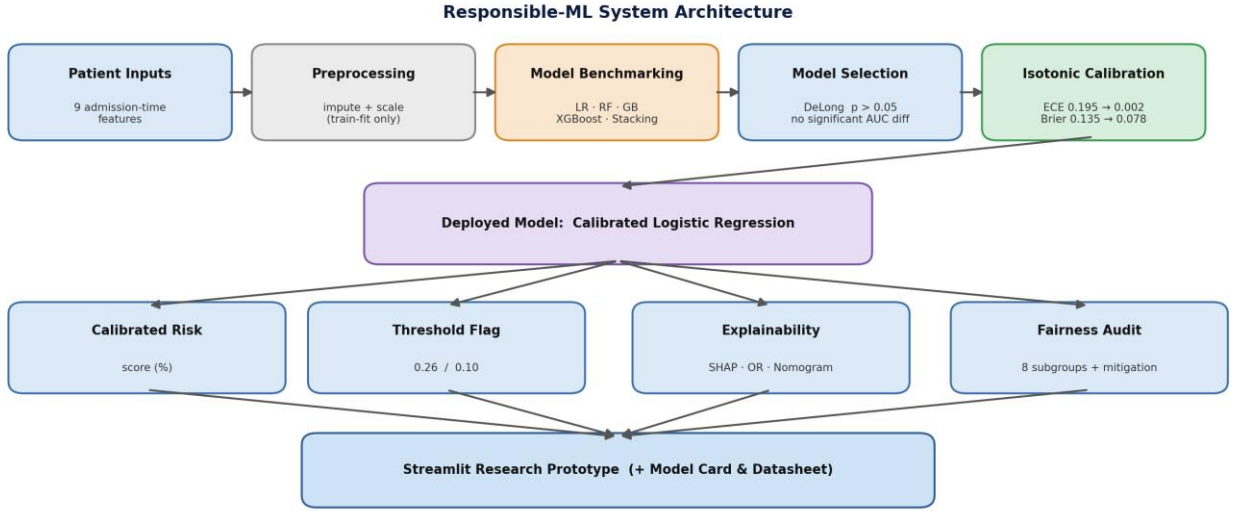


Fig. 1. Responsible-ML system architecture. Nine admission-time features are preprocessed leakage-free and benchmarked across five model families. The deployed model is selected by DeLong test and isotonic-calibrated, then emits a calibrated risk score, a tuned threshold flag, multi-layer explanations, and a subgroup fairness audit, all exposed through a Streamlit prototype.

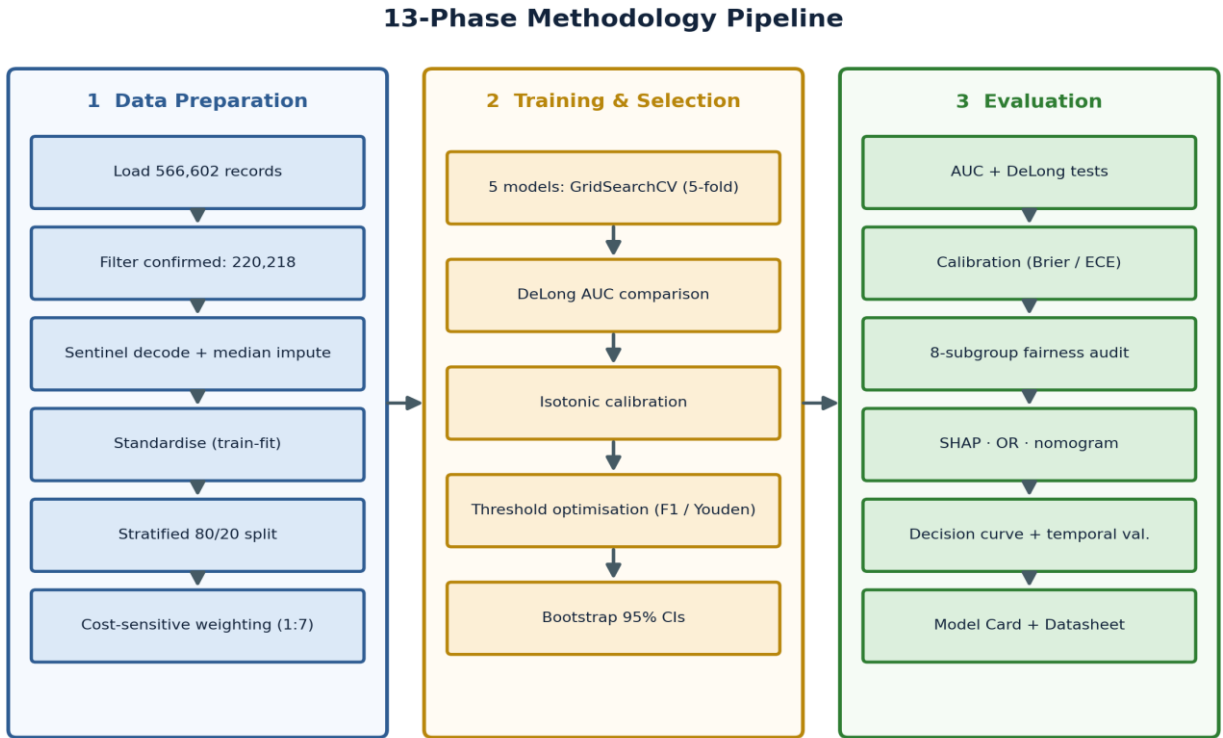


Fig. 2. 13-phase methodology pipeline. Phase 1 (data preparation) applies leakage-free preprocessing and cost-sensitive sampling; Phase 2 (training and selection) benchmarks five models, tests AUC differences, and calibrates the deployed model; Phase 3 (evaluation) computes discrimination, calibration, fairness, explainability, decision-curve, and temporal results, and emits governance artefacts.

A. Dataset and Cohort Selection

We use the Mexican Government COVID-19 surveillance dataset on Kaggle [19] (566,602 records for the acute phase). Each record includes age, sex, comorbidity flags (diabetes, hypertension, cardiovascular disease, pneumonia, obesity, asthma, COPD), outcome fields (intubation, ICU admission, date of death) and administrative variables. We restrict to confirmed cases ($\text{covid_res} = 1$), $n = 220,218$. Of these, 27,102 died in hospital and 193,116 survived - a $\sim 7:1$ imbalance and 12.31% mortality. Data are split 80/20 stratified by outcome (176,174 train / 44,044 held-out test). The events-per-variable ratio (EPV = 3,011) is far above the recommended

minimum of 10, so coefficient overfitting is not a concern. Table II summarises the cohort.

TABLE II
Dataset and Cohort Composition

Property	Value
Source	Mexican Govt. COVID-19 surveillance [19]
Total records	566,602
Confirmed cohort	220,218 ($\text{covid_res} = 1$)
Outcome / prevalence	in-hospital mortality / 12.31%
Class imbalance	1 : 7 (27,102 vs 193,116)
Features (9)	age, sex, 7 binary comorbidities
Train / Test	176,174 / 44,044 (stratified 80/20)
Events-per-variable	3,011 (> 10)

B. Preprocessing and Leakage Control

The outcome label was built from date_died: any date other than '9999-99-99' -> 1 (deceased), else 0. Administrative comorbidity encodings (1 = Yes, 2 = No) were recoded to binary (1.0/0.0). Values 97/98/99 (unknown/not applicable) were replaced with NaN before imputation, affecting 0.3% of comorbidity entries. SimpleImputer and StandardScaler were fitted on the training partition and applied to the held-out partition, all inside sklearn pipelines so the split-then-fit paradigm holds within each CV fold. Intubation status and ICU admission were excluded from the outset. The nine admission-time inputs are age, sex, diabetes, hypertension, cardiovascular disease, pneumonia, obesity, asthma and COPD.

C. Statistical Feature Validation

We validated each candidate feature against mortality before training. Age was significantly higher in decedents (independent-samples t-test, 61.0 vs 43.6 years; $t = 176.99$, $p < 0.001$). Chi-square tests showed each of seven binary comorbidities was significantly related to the outcome (all $p < 0.001$). Benjamini-Hochberg FDR correction over five main tests was applied and all passed ($q < 0.05$). The maximum pairwise Pearson correlation was 0.34, well below any multicollinearity threshold. The final nine features are statistically validated, compact and non-redundant [20].

D. Model Training and Hyperparameter Tuning

Five models were trained: Logistic Regression (L2), Random Forest, Gradient Boosting, XGBoost, and a stacking ensemble (LR/RF/GB base learners, LR meta-learner). Cost-sensitive learning was used throughout: class_weight='balanced' for LR and RF, sample_weight for Gradient Boosting, scale_pos_weight = 7 for XGBoost. A 5-fold stratified GridSearchCV on training data only returned LR C = 0.01 (CV AUC 0.891 +/- 0.002) and RF max_depth = 6, n_estimators = 200. We also examined SMOTE: it matched the test AUC (0.888) while adding 132,810 synthetic patients and 38% training time, so the lighter cost-sensitive option was kept. random_state = 42 throughout for reproducibility, and 44 unit tests in test_analysis.py verify all statistical helper functions independently before any model is trained. The Logistic Regression models the standardised log-odds of mortality (Eq. 1). Table III lists the tuned hyperparameters.

$$P(y=1 | x) = \sigma(\beta \cdot x + \beta_0), \quad \sigma(z) = 1 / (1 + e^{-z}) \quad (1)$$

TABLE III
Tuned Hyperparameters

Hyperparameter	Value
Optimiser / solver	GridSearchCV (5-fold stratified)
LR regularisation C	0.01 (L2)
RF depth / trees	6 / 200
XGBoost	7
scale_pos_weight	
Class-imbalance handling	class_weight / sample_weight / scale_pos_weight
Calibration	isotonic, CalibratedClassifierCV (cv=5)
Bootstrap resamples (CI)	2,000
Random state	42 (reproducible)

E. Probability Calibration

The pre-calibration reliability diagram showed systematic overconfidence across the probability scale (Brier = 0.135, ECE = 0.195). Deviations were non-monotonic, excluding Platt scaling (which assumes monotone sigmoid distortion) and favouring isotonic regression. The deployed

model is therefore a CalibratedClassifierCV (method='isotonic', cv=5) over the tuned Logistic Regression, saved as model_primary_deployed.joblib (13.0 KB). For model selection we used DeLong's method [21] to compare held-out AUCs across all five models, with 2,000-resample bootstrap CIs. Calibration quality is quantified by the Brier score (Eq. 2) and count-weighted ECE over M bins (Eq. 3).

$$BS = (1/N) \sum (p_i - y_i)^2 \quad (2)$$

$$ECE = \sum (|B_m| / N) \cdot |\text{acc}(B_m) - \text{conf}(B_m)| \quad (3)$$

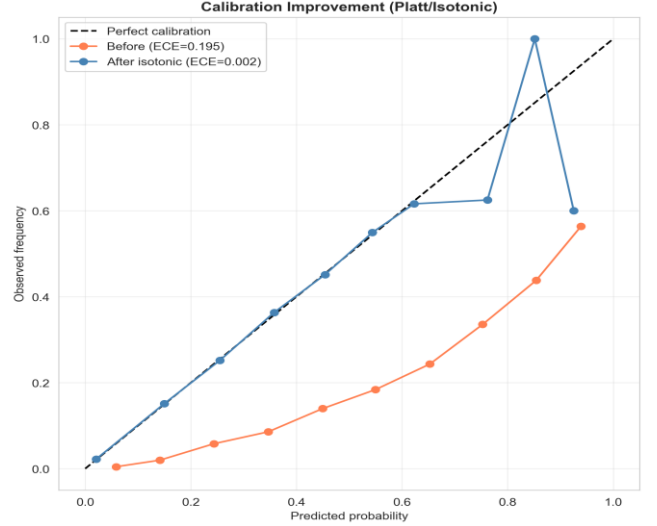


Fig. 3. Reliability diagram before and after isotonic calibration. Post-calibration probabilities closely track observed mortality rates across all risk deciles.

F. Multi-Layer Explainability Framework

Three explainability layers are provided, each for a different audience, all scoped to the model that actually makes the prediction. On the technical layer, SHAP values are computed on the deployed Logistic Regression (LinearExplainer), so the attributions describe the same model that produces each patient's risk; a secondary TreeSHAP [22] check on an XGBoost model gives global rankings and per-patient waterfall plots for comparison only. An unregularised statsmodels Logit provides odds ratios per +1 SD with 95% Wald intervals in a format clinicians already know; these are association estimates reported separately from the deployed regularised (C=0.01) model's shrunk coefficients. Integer point scores from log-odds coefficients form a nomogram usable on paper for the point-of-care layer without software. As a secondary, exploratory check on the tree model (the deployed Logistic Regression is additive and models no interactions), the strongest pairwise interaction was pneumonia x age (0.182), indicating that elderly pneumonia patients face higher risk of acute respiratory involvement. The odds ratio is the exponent of the logistic coefficient (Eq. 4), i.e., the multiplicative change in mortality odds per +1 SD.

$$OR_j = \exp(\beta_j) \quad (4)$$

G. Subgroup Fairness Audit and Threshold Mitigation

The audit covers eight pre-specified subgroups (age 18-30, 30-50, 50-65, 65+; female vs male; diabetes present vs absent) and two measures each: AUC (discrimination fairness) and count-weighted ECE (calibration equity). A subgroup is flagged when its AUC is outside five percentage points of the overall AUC. Where flagged, group-specific thresholds equalising true-positive rates are computed. This

mitigation, without retraining, reduced TPR disparity from 0.603 to 0.045. Resulting thresholds: age 18-30 $t=0.05$, 30-50 $t=0.12$, 50-65 $t=0.23$, 65+ $t=0.41$.

H. Web Prototype Architecture

The prototype is a two-page Streamlit application that loads the same three pre-fitted joblib artefacts used in training (imputer, scaler, calibrated model) and uses the identical impute -> scale -> predict_proba chain. Inputs are eight comorbidity toggles and an age slider. Outputs are the calibrated risk percentage in four colour bands (LOW/MEDIUM/HIGH/CRITICAL), per-patient odds-ratio attribution bars, a global SHAP panel, and the nomogram point total. Performance figures are read from exported CSVs at runtime, so displayed values always reflect the actual model. Four disclaimer notices frame all outputs as mortality risk estimates, not validated estimates for individual clinical decisions.

I. Temporal Out-of-Time Validation

An out-of-time split by admission date tests temporal robustness: training on records before 15 June 2020 (176,979

patients) and validating on records from 15 June 2020 onward (43,239 patients). This is harder than a random split because clinical protocols and case mix changed over the period, and the model must generalise across that shift.

J. Baseline Model

Two baselines contextualise results. A majority classifier (predict survival) reaches 0.877 accuracy but AUC 0.50 and recall 0.0 - illustrating why accuracy misleads at <12% prevalence and why we report AUC, calibration and net benefit. The second baseline is a plain, unweighted, uncalibrated Logistic Regression at threshold 0.5, isolating the contributions of cost-sensitive weighting, isotonic calibration and threshold optimisation. Relative to it, the deployed model has far lower ECE (0.195 -> 0.002) and higher operating-point sensitivity (0.296 -> 0.702) at the same AUC (Table IV).

IV. PERFORMANCE EVALUATION

A. Model Discrimination and DeLong Testing

TABLE IV

Full Model Benchmark, Test Set ($n = 44,044$). Sens.=Sensitivity, Spec.=Specificity, Prec.=Precision, ECE=count-weighted Expected Calibration Error. Bold = deployed model.

Model	AUC	95% CI	Sens.	Spec.	Prec.	F1	Brier	ECE	DeLong vs LR	Deploy
Logistic Regression	0.888	0.884-0.892	0.662	0.892	0.463	0.545	0.078	0.002	Reference	Yes
Random Forest	0.866	0.861-0.871	0.673	0.876	0.432	0.526	0.081	0.006	$p \sim 3e-42$	No
Gradient Boosting	0.888	0.884-0.892	0.716	0.873	0.441	0.546	0.078	0.004	$p=0.172$ n.s.	No
XGBoost	0.888	0.884-0.892	0.656	0.895	0.466	0.545	0.078	0.004	$p=0.594$ n.s.	No
Stacking Ensemble	0.888	0.884-0.892	0.677	0.887	0.457	0.546	0.078	0.003	$p=0.243$ n.s.	No

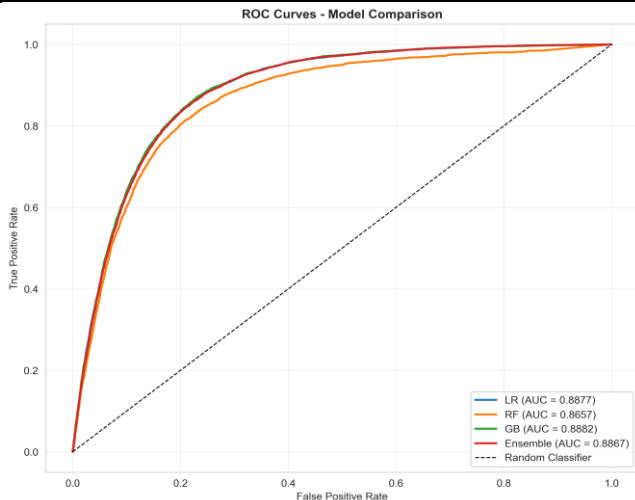


Fig. 4. ROC curves for all five model families. Calibrated LR shows no significant AUC difference from the ensembles by DeLong test; RF is significantly worse ($p \sim 3e-42$).

The four remaining models are statistically indistinguishable except Random Forest, which is significantly worse than Logistic Regression (DeLong $p \sim 3e-42$). Gradient Boosting, XGBoost and Stacking are all statistically equivalent to LR at AUC 0.888. Five-fold CV gave AUC 0.891 $\pm$ 0.002, matching the temporal out-of-time AUC (0.891). This resolves RQ2: no complex model performs significantly better than the interpretable Logistic Regression, and the discrimination cost of the simpler model is none. A non-significant DeLong result indicates no detectable AUC difference, not formal non-inferiority (which would require a pre-specified margin); the

interpretable model is therefore preferred on parsimony, calibration and transparency grounds.

B. Calibration Analysis

TABLE V

Calibration Metrics Before and After Isotonic Correction

Metric	Pre-calibration	Post-isotonic
Brier Score	0.135	0.078
ECE (count-weighted)	0.195	0.002
ECE (bin-averaged)	0.287	0.064

Isotonic calibration improves all three metrics substantially. Count-weighted ECE of 0.002 means predicted probabilities and observed rates are nearly matched across the population; the bin-averaged ECE of 0.064 confirms good behaviour even in rare high-risk bins. A predicted rate of 0.30 corresponds to roughly 30% mortality in the held-out set. The output is well-calibrated on this held-out cohort.

C. Threshold Optimisation

TABLE VI

Threshold Optimisation Strategies

Strategy	Threshold	Sensitivity	Specificity	Precision	F1
Default (0.5)	0.50	0.296	0.968	0.564	0.388
F1-optimal	0.26	0.702	0.878	0.447	0.546
Youden's J	0.10	0.877	0.761	0.340	0.490

For clinical triage we recommend the F1-optimal threshold of 0.26 (sensitivity 0.702, precision 0.447 - 3.6x the 12.3% base rate). For population screening, where sensitivity matters more, Youden's J = 0.10 raises sensitivity to 0.877 at lower specificity.

D. Feature Importance and Explainability

TABLE VII

Deployed-Model (Logistic Regression) SHAP Importance and Odds Ratios (statsmodels, per +1 SD)

Feature	SHAP %	OR (95% CI)	Sig.
Pneumonia	39.9	2.65 (2.61-2.69)	***
Age	33.1	2.33 (2.28-2.37)	***
Sex (female)	10.9	0.80 (0.79-0.82)	*** protective
Diabetes	6.9	1.15 (1.13-1.17)	***
Hypertension	4.3	1.09 (1.07-1.10)	***
Obesity	3.7	1.10 (1.08-1.12)	***
COPD	0.6	1.03 (1.02-1.04)	***
Asthma	0.3	0.98 (0.97-1.00)	n.s.
Cardiovascular	0.2	1.00 (0.99-1.01)	n.s.

Significance of the odds ratio: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. = not significant ($p \geq 0.05$). SHAP % are from the deployed Logistic Regression; odds ratios from an unregularised statsmodels Logit.

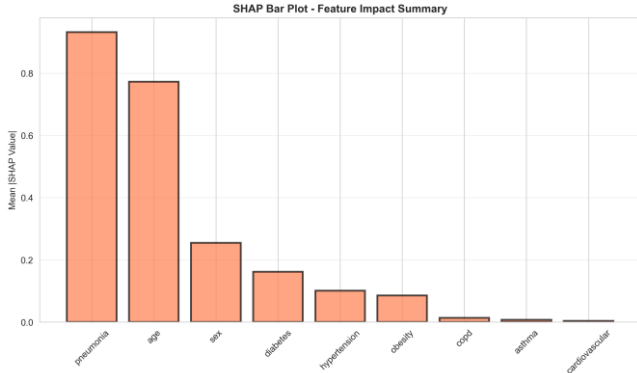


Fig. 5. Global feature importance. Pneumonia and age dominate the ranking, together carrying most of the importance and indicating a concentrated, largely linear signal; deployed-model (Logistic Regression) SHAP percentages are reported in Table VII.

Pneumonia is the dominant - and most modifiable - risk factor (OR 2.65, SHAP 39.9%), followed by age as the strongest non-modifiable predictor (OR 2.33, SHAP 33.1%); together they carry 73.0% of the deployed model's SHAP importance. Female sex is protective (OR 0.80), consistent with the broader COVID-19 literature, while cardiovascular disease and asthma contribute essentially nothing (cardiovascular 0.2%, asthma 0.3% SHAP; OR ~ 1.00 , $p > 0.05$) - reported as a deliberate null result. The strongest pairwise interaction from the exploratory tree-SHAP check, pneumonia x age (0.182), identifies older patients with active respiratory involvement as highest risk. As a point-of-care reference, the nomogram assigns: pneumonia +24, age +6/decade, diabetes +5, COPD +4, hypertension +3, obesity +3, cardiovascular +1, asthma -1, female -5.

E. Subgroup Fairness Audit Results

TABLE VIII

Subgroup Fairness Audit (FLAG = AUC deviation > 5pp from overall)

Subgroup	n	AUC	Flag	ECE
Age 18-30	6,290	0.891	OK	0.002
Age 30-50	19,713	0.874	OK	0.004
Age 50-65	11,189	0.799	FLAG	0.008
Age 65+	5,926	0.733	FLAG	0.015
Female	19,961	0.901	OK	0.005
Male	24,083	0.873	OK	0.003
Diabetes	7,160	0.776	FLAG	0.012
No Diabetes	36,704	0.899	OK	0.004

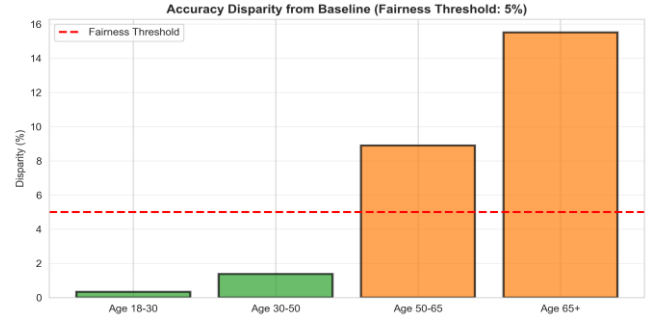


Fig. 6. Subgroup fairness audit: AUC and ECE across eight groups. ECE < 0.015 in all subgroups; AUC drops in older and diabetic subgroups reflect restriction of range.

Three subgroups are flagged on discrimination: age 50-65 (AUC 0.799), 65+ (AUC 0.733), and diabetes (AUC 0.776). But ECE remains low in all eight subgroups - the more clinically relevant result - showing well-calibrated risk estimates across groups. The gender AUC gap is 2.8%, near the 5% flag threshold. To address RQ3, group-specific threshold mitigation reduced TPR disparity from 0.603 to 0.045.

F. Decision Curve Analysis and Sensitivity Analysis

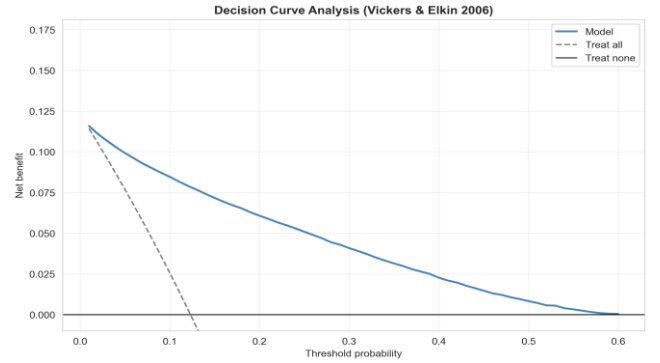


Fig. 7. Decision Curve Analysis. Model net benefit (0.085 at threshold 0.10) substantially exceeds treat-all (0.026) and treat-none across the clinical range.

Decision Curve Analysis confirms positive net benefit across the clinically relevant threshold range (0.05-0.50). At threshold 0.10 the net benefit is 0.085, exceeding treat-all (0.026), confirming utility beyond accuracy. Counterfactual sensitivity analysis reveals that removing pneumonia from a patient's profile produces the biggest individual risk reduction - in one illustrative example predicted risk dropped from 42.0% to 8.0% (delta = -34.0 pp), consistent with its high SHAP importance. Across the test set, risk distributes as LOW <30% (83.3%), MEDIUM 30-50% (10.3%), HIGH 50-70% (6.4%), CRITICAL $\geq 70\%$ (0.05%).

G. Ablation: Contribution of Pipeline Components

To isolate each stage's contribution, Table IX builds the deployed model additively from a plain logistic baseline. Cost-sensitive weighting (class_weight='balanced') is the training-time imbalance handler that yields the 0.888 AUC under the 1:7 ratio - matched exactly by SMOTE at 0.888 (Section III-D). On the weighted model, isotonic calibration then cuts Brier from 0.135 to 0.078 and count-weighted ECE from 0.195 to 0.002 without changing discrimination, and threshold optimisation (0.26) raises operating-point sensitivity from 0.296 to 0.702. Because calibration and thresholding preserve the model's ranking, AUC is unchanged throughout; each stage improves a distinct, non-overlapping property.

TABLE IX

Ablation: Cumulative Contribution of Pipeline Components (deployed Logistic Regression).

Configuration	AUC	Brier	ECE	Sens.	Spec.
Baseline LR (uncalibrated, thr 0.5)	0.888	0.135	0.195	0.296	0.968
+ Isotonic calibration	0.888	0.078	0.002	0.296	0.968
+ Threshold optimisation (0.26) = deployed	0.888	0.078	0.002	0.702	0.878

AUC is invariant because calibration and thresholding preserve ranking. Cost-sensitive weighting - the imbalance handler underlying the 0.888 AUC (SMOTE-matched, Section III-D) - improves minority-class learning rather than the 0.5 operating point, so it is credited in the text.

V. DISCUSSION

A. Why Logistic Regression Ties the Ensembles

None of the more complex models outperforms Logistic Regression, and the SHAP results explain why: pneumonia and age together account for 73.0% of feature importance, both increase monotonically, and there are no significant interactions. When two features dominate so completely and their effects are roughly linear and additive, tree ensembles have no higher-order or non-monotonic structure to exploit and instead overfit noise; our DeLong test is consistent with this on 220,218 patients, a more rigorous comparison than Bjerre et al. [8]. Two further criteria favour LR for deployment. First, calibration: Table IV shows LR achieves the lowest ECE (0.002), better than every ensemble (RF 0.006, GB 0.004, XGBoost 0.004, Stacking 0.003). Second, interpretability: LR gives native odds ratios, linear SHAP attribution and a points-based nomogram usable at the point of care without software - none of which the ensembles match while offering no better AUC or Brier score. Across discrimination, calibration and interpretability, Logistic Regression is the best deployment choice.

B. Calibration as the Primary Deployment Criterion

The pre-calibration ECE of 0.195 shows how misleading an uncalibrated model can be: predicted probabilities diverged from observed mortality rates by nearly 20 percentage points on average. After isotonic calibration, ECE fell to 0.002 (count-weighted) - near the minimum possible - so the model functions as a reliable probability estimator on this cohort. That risk estimates are not systematically biased across the eight subgroups is equally significant: a minimum requirement for equitable clinical AI.

C. Interpreting Subgroup AUC Drops

The lower AUCs (0.733 for age 65+, 0.776 for diabetes) look worrisome but arise from statistical causes. In a homogeneously high-risk subgroup (older or diabetic patients), rank variation is compressed, which lowers AUC even when calibration holds - visible in the calibration results (ECE 0.015 for age 65+, 0.012 for diabetes, both acceptable). The model gives these groups precise risk estimates but less room to rank them. A key limitation in the clinical-AI literature is conflating discrimination fairness with calibration fairness; when AUC declines across subgroups it is too often reported as algorithmic bias without checking whether calibration was maintained.

D. Responsible AI and Governance Framing

Responsible-AI practices are built into the pipeline at three levels rather than added afterwards. At the model level: leakage-free split-then-fit design, empirical calibration

verification, two use-case threshold strategies, and temporal (not random) validation. At the fairness level: eight subgroups audited on two lenses, transparent reporting of null and flagged results, and group-specific mitigation cutting TPR disparity to 0.045. At the documentation level: a Model Card [23], a Datasheet [24], and four in-app disclaimers framing outputs as mortality-risk estimates.

E. Limitations and Threats to Validity

Several limitations bound the conclusions. The most important is that the outcome is in-hospital mortality, not Long COVID: the two are distinct outcomes (a patient who dies cannot subsequently develop Long COVID), so this is a COVID-19 mortality prediction study, not a Long COVID predictor. Predicting Long COVID directly would require PASC-labelled follow-up data, which no large-scale public dataset currently provides. On external validity, the data are from a single country (Mexico) and a single early-pandemic phase (March-November 2020), before vaccine rollout and major variants; replication in other health systems, populations and later phases is unknown. The feature set is limited to 9 admission-time variables without laboratory values, vital signs, comorbidity severity or variant data. Finally, the temporal out-of-time test, though harder than a random split, is still single-nation; there has been no multi-site or multi-country external validation.

VI. CONCLUSION

We present a 13-phase end-to-end responsible-ML pipeline for COVID-19 in-hospital mortality risk that can serve as a reusable framework for severe-outcome prediction. The main empirical finding is a formal DeLong comparison ($p > 0.05$) between a calibrated, interpretable Logistic Regression and Random Forest, Gradient Boosting, XGBoost and a Stacking ensemble on 44,044 held-out patients, showing no statistically significant AUC difference for the LR model at an AUC of 0.888 (95% CI 0.884-0.892). SHAP analysis explains this: pneumonia and age contribute 73.0% of the signal, which is essentially linear with only shallow pairwise interactions - precisely why complex models offer no advantage over interpretable regression.

The key next steps are to apply the pipeline to PASC-labelled cohorts such as RECOVER [10] or UK Biobank, where the label reflects Long COVID directly rather than mortality; to add longitudinal biomarkers, vaccination history and variant data; to perform multi-site external validation before any clinical use; and to test the prototype prospectively with a clinical outcome measure.

ACKNOWLEDGMENT

I thank Dr Martin Crane and Dr Tai Tan Mai (School of Computing, Dublin City University) for their supervision, feedback and guidance. The Mexican Government COVID-19 surveillance dataset was obtained from Kaggle [19]; I thank the public-health authorities that provided it. This work was undertaken for the M.Sc. in Computing (Data Analytics) at Dublin City University, Ireland.

GENERATIVE AI DISCLOSURE

I used a generative-AI assistant (Anthropic's Claude) during this project in three ways: to help review and debug my Python analysis pipeline, to help edit sections of this report from the numerical results my own code produced, and for IEEE formatting and layout. All data processing, model training, statistical tests and results were produced by

running my own code, and I checked every table, figure and number in this paper against my pipeline's outputs. AI also contributed methodological and editorial suggestions, which I evaluated, accepted or rejected, and adapted. I reviewed and revised all AI-assisted text, and I take full responsibility for the final content.

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